EVERY PROPER SURJECTIVE MORPHISM ONTO AN INTEGRAL ALGEBRAIC SPACE CONTAINS AN ALTERATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-over-fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the choice of a closed point in the fibre over the generic point of an open affine of the target, the verification of surjectivity, properness and the singleton fibre of the restriction to the closure, and the conclusion through the characterization of alterations and the inheritance of decency.

Editorial note. The label lemma-alteration-contained-in names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-over-fields.tex –label lemma-alteration-contained-in –start 1413 –end 1446 –local-id 28062 returned RESULT=OK (L28062 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-over-fields/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Existence of a nonempty open affine of a decent space. Decent Spaces, the theorem producing a dense open subscheme, applied to obtain a nonempty open affine V of the integral target. Supplies the open affine V and its generic point eta used to pick a closed point of the fibre.
- Existence of a closed point in a quasi-compact sober space. Topology, the lemma that a quasi-compact sober space has a closed point, together with the description of the fibre of a proper morphism as a quasi-compact sober space. Supplies the closed point x of the fibre over the generic point.
- Reduced induced closed subspace structure. Properties of Spaces, the definition of the reduced induced closed subspace structure on the closure of a point. Supplies the closed subspace X' whose underlying space is the closure of x.
- Characterization of alterations and decency of the source. Same chapter, the lemma characterizing an alteration by properness, surjectivity and generic finiteness, and Decent Spaces, the lemmas that decency is inherited by a separated space over a decent one. Supply the conclusion that the restriction is an alteration.

Date: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03M5: sources/stacks-chow/tags.txt, source lines 4737–4737. Prerequisite pointer invoked at proof line(s) 1442; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ABY: sources/stacks-chow/tags.txt, source lines 13036–13036. Prerequisite pointer invoked at proof line(s) 1443; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AC8: sources/stacks-chow/tags.txt, source lines 13046–13046. Prerequisite pointer invoked at proof line(s) 1438; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 03K6: sources/stacks-chow/tags.txt, source lines 4666–4666. Prerequisite pointer invoked at proof line(s) 1429; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 086U: sources/stacks-chow/tags.txt, source lines 10404–10404. Prerequisite pointer invoked at proof line(s) 1423; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ADB: sources/stacks-spaces-over-fields/pinned-original-source.tex, source lines 1402–1402. Prerequisite pointer invoked at proof line(s) 1445; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 047X: sources/stacks-chow/tags.txt, source lines 5524–5524. Prerequisite pointer invoked at proof line(s) 1433; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 005E: sources/stacks-chow/tags.txt, source lines 206–206. Prerequisite pointer invoked at proof line(s) 1430; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper surjective morphism of algebraic spaces over S. Assume Y is integral. Then there exists an integral closed subspace $X' \subset X$ such that $f' = f|_{X'} : X' \to Y$ is an alteration.*

Proof. Let $V \subset Y$ be a nonempty open affine (Decent Spaces, Theorem 086U). Let $\eta \in V$ be the generic point. Then X_η is a nonempty proper algebraic space over η. Choose a closed point $x \in |X_\eta|$ (exists because $|X_\eta|$ is a quasi-compact, sober topological space, see Decent Spaces, Proposition 03K6 and Topology, Lemma 005E.) Let X' be the reduced induced closed subspace structure on $\overline{\{x\}} \subset |X|$ (Properties of Spaces, Definition 047X. Then $f' : X' \to Y$ is surjective as the image contains η. Also f' is proper as a composition of a closed immersion and a proper morphism. Finally, the fibre X'_η has a single point; to see this use Decent Spaces, Lemma 0AC8 for both $X \to Y$ and $X' \to Y$ and the point η. Since Y is decent and $X' \to Y$ is separated we see that X' is decent (Decent Spaces, Lemmas 03M5 and 0ABY). Thus f' is an alteration by Lemma 0ADB. □